



Uncertainty Quantification in Stochastic Pathogen Spread for Precision Agricultural Decision Support

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Abstract: The spatiotemporal spread of plant pathogens introduces substantial uncertainty into agricultural production systems, thereby complicating timely disease management and resource allocation. An information-theoretic framework was developed to quantify structural uncertainty in pathogen transmission by integrating stochastic population dynamics with system entropy. A finite agricultural field containing a homogeneous crop population was represented as a two-state system, in which individual plants existed in either a healthy or an infected state. Pathogen transmission was modeled as a birth-death process, while the probability distribution of the infection state was characterized using a binomial formulation under the assumed population structure. On this basis, system entropy was introduced as a quantitative measure of structural uncertainty. It was demonstrated that entropy followed a characteristic inverted-U-shaped trajectory, increasing during the early stages of disease propagation and reaching a maximum at an intermediate level of infection intensity, where uncertainty and system volatility were greatest. As pathogen spread approached saturation, entropy progressively decreased. This entropy maximum was shown to define a critical operational threshold at which surveillance, treatment, and resource deployment can be implemented with the greatest expected effectiveness. Beyond operational decision support, the proposed framework establishes a quantitative basis for evaluating long-term risk mitigation strategies. Reductions in entropy achieved through the deployment of disease-resistant crop varieties, improved biosecurity measures, optimized field infrastructure, or enhanced monitoring systems can be directly interpreted as reductions in structural uncertainty, thereby providing measurable indicators for minimizing disease-related economic losses, improving intervention efficiency, and strengthening the resilience of agricultural production systems and supply chains. By establishing a rigorous connection between stochastic epidemiological dynamics and information theory, the proposed framework provides a generalizable analytical foundation for uncertainty-aware agricultural management and data-driven decision-making under pathogen risk.

Keywords: Information and entropy; Population dynamics; Decision support; Agriculture; Binomial system

1 Introduction

Regarding field precision spraying, instead of blanket-spraying an entire greenhouse facility, Internet of Things sensors or multi-spectral drone cameras can calculate the real-time sample infection rate for individual zones of plants in a given field. The decision rule would be to apply direct chemical applications specifically to zones operating approximately at the maximum entropy. And operational savings would be that spraying low-entropy zones is either a waste of chemical assets (if the infection rate is very small), or a waste of capital on dead plants (if the infection rate is very large). Targeting the high-entropy tipping point reduces fungicide volume, saving the maximum possible yield. Labor is one of the highest operational costs in greenhouse farming. Regarding dynamic scouting schedules, when a zone enters day 20 to 35 (i.e., the high-slope entropy acceleration window) during a two-month period, operational dashboards automatically flag it for daily human scouting. Once a zone passes day 45, and enters asymptotic absorption (i.e., entropy-decaying period), human scouting is turned off, and labor is reallocated elsewhere because the outcome is already highly certain.

Stochastic pure-birth processes, which model populations experiencing only growth, are well-documented in foundational texts [1–4]. Similarly, compartment-based epidemic models are widely utilized to simulate the spread

of infectious diseases [5–7]. Building on these frameworks, recent advancements in the mathematical modeling of population dynamics and infection transmission have been extensively analyzed [8–13]. This current study aims to bridge pure analytical modeling with agribusiness risk by demonstrating that the uncertainty associated with a given field acts as a quantitative measure on operational overhead. Agronomists can dynamically mitigate the disease by using entropy-driven thresholds to trigger precise, localized interventions before a plant field crosses the information-based point of no return.

Recent advancements in precision agriculture leverage a combination of remote sensing, IoT infrastructure, and machine learning to improve crop health management and sustainability. For instance, unmanned aerial vehicle (UAV)-based multispectral imaging has proven effective in monitoring canopy vigor and crop rotation outcomes [14], and the growing importance of integrating advanced sensing technologies, such as remote sensing and IoT, for real-time monitoring and data-driven decision-making is highlighted in the study [15]. The integration of the IoT data streams with machine learning algorithms—particularly convolutional neural networks—has demonstrated high accuracy in the early detection and classification of plant diseases [16]. Furthermore, these technological synergies enable targeted pesticide application strategies that can potentially reduce chemical inputs by up to 90% [17].

This current study investigates invasive pest management, considering some fields across a given country. In each field, M plants are randomly tested for the presence of a destructive beetle. The number of infected plants N , with $N \in \{0, 1, 2, \dots, M\}$, in any given field follows a binomial distribution: $N \sim \text{Bin}(M, p)$, where p is the infection rate of that specific field. There are three limit cases: i) Clean field, with 0 out of M plants infected, meaning perfect certainty, and the pest has not established itself; ii) Epidemic field, with M out of M plants infected, meaning perfect certainty again, and the field is completely overrun; iii) Frontline field, with $M/2$ out of M plants infected, meaning maximum uncertainty, and this field is the “battleground” where the pest is actively spreading, but has not won yet. Instead of just mapping where the pest is, this study aims to map where the pest manifests the most uncertainty. When entropy $S \approx S_{max}$ (i.e., high-entropy fields), where S_{max} denotes the maximum entropy of the field, these are the ecological tipping points. Therefore, farmers should target these fields with immediate intervention, because they represent active, volatile outbreak zones. When entropy $S \approx 0$ (i.e., low-entropy fields), this is either safe (for now), or a lost cause, requiring less dynamic monitoring. In this setup, a field is modeled as a closed, finite system where plants can exist in one of two states: healthy or infected. The spread of a pathogen within that finite space is modeled as a binomial birth-death process [18]. This study then tracks the entropy of the system’s transition state to locate the exact point of the maximum biological chaos.

In this work, depending on the concepts of information i and entropy $S = E(i)$ associated with an agricultural field of crop plants, with a current number N of infected plants, and a maximum number M of plants, and modeling the field by a birth-death process of size $M + 1$ and states n , with $n = 0, 1, 2, \dots, M$, this study promotes a stochastic analysis of agricultural population dynamics. The information i possessed by a field (i.e., carried out by the random variable N) is an essential characteristic of the distinct states of the field (i.e., of the birth-death process) and, by definition, the entropy $S = E(i)$ of the field is the uncertainty associated with the field. This study adopts the Gibbs’ entropy formula [19] and its formulation in terms of random variables, proposed initially by Shannon [20] (focusing only on the discrete case).

The remainder of this study is structured as follows. Section 2 reports the basic model and introduces the performance measures in the methodology. Section 3 considers a case study, modeling the pathogen spread in a greenhouse tomato plot. Section 4 provides the conclusion of this study.

2 System Model

It is assumed that an agricultural field contains a fixed population of M homogenous crop individuals. The state of the field is characterized by the random variable N , with $N \in \{0, 1, 2, \dots, M\}$, representing the total number of infected plants. The number of remaining susceptible hosts is $M - N$. This study models the transmission and recovery dynamics as a finite-space birth-death process with states n ($n = 0, 1, 2, \dots, M$). Let λ denote the birth rate (that is, the rate at which an infected host infects a susceptible host) and μ denote the death rate (that is, the rate at which an infected host recovers, or is removed). Then, $\rho = \lambda/\mu$ is the field infection intensity parameter. Furthermore, the transition rate λ_n is the plant infection rate given n infected plants (at transition $n \rightarrow n + 1$), and μ_{n+1} is the plant recovery rate given $n + 1$ infected plants (at transition $n + 1 \rightarrow n$).

2.1 Basics of the Model

The quantity N , which defines the specific states of the field, is a random variable with a probability distribution $p_n = \text{Prob}\{N = n\}$, with $n = 0, 1, 2, \dots, M$. This distribution is governed by the probability continuity law (via the global balance equations) and the probability conservation law (via the normalization condition) [21–23].

Consequently, the probability distribution p_n , with $n = 0, 1, 2, \dots, M$, is determined by the following relations:

$$\frac{p_n(\rho)}{p_0(\rho)} = \prod_{j=0}^{n-1} \frac{\lambda_j}{\mu_{j+1}} = a_n \cdot \rho^n, n = 0, 1, 2, \dots, M,$$

$$p_0(\rho) = \frac{1}{f_M(\rho)}, \quad f_M(\rho) = 1 + \sum_{n=1}^M a_n \cdot \rho^n, \quad \rho > 0, \quad (1)$$

where, the coefficients $a_n > 0$, do not depend on the parameter ρ .

Thus, this study obtains a family of birth-death processes of size $M + 1$, indexed by the parameter ρ , which models the plant infection/recovery mode, i.e., it models the family of fields, indexed by the parameter ρ and associated with the polynomial $f_M(\rho)$.

The quantity N denotes the current number of infected plants in such a family of fields. Therefore, for a given field of size M , any value ρ' of the parameter ρ defines a (equilibrium) macrostate of the field. The parameter ρ , being a ratio of frequencies, has a kinematic nature. Being in any of its specific states n in an equilibrium macrostate, a given field possesses a quantity of information $i_N(\rho)$ with possible values $i_n(\rho)$, and its expected value, i.e., entropy $S(\rho)$, as follows:

$$i_n(\rho) \stackrel{\text{def}}{=} -\ln p_n(\rho), n = 0, 1, \dots, M, \rho > 0, \quad (2)$$

$$S(\rho) = E(i(\rho)) \stackrel{\text{def}}{=} \sum_{n=0}^M i_n(\rho) \cdot p_n(\rho), \quad \rho > 0. \quad (3)$$

The mean value of the variable N , which represents the average number of infected plants within a specified field, i.e., the function $\bar{N}(\rho)$, is given by:

$$\bar{N}(\rho) \stackrel{\text{def}}{=} \sum_{n=0}^M n \cdot p_n(\rho) = \rho \cdot \frac{f'_M(\rho)}{f_M(\rho)} = \rho \cdot [\ln f_M(\rho)]'_\rho = \rho \cdot [i_0(\rho)]'_\rho, \rho > 0. \quad (4)$$

The derivative of $\bar{N}(\rho)$ with respect to the parameter ρ is ascertained as:

$$\rho \cdot [\bar{N}(\rho)]'_\rho = \text{Var}(N_{f_M(\rho)}) > 0, \rho > 0. \quad (5)$$

Therefore, the function $\bar{N}(\rho)$ is a monotonically increasing function of the infection intensity parameter ρ . The entropy, by definition, represents the uncertainty associated with the field. The function $S(\rho) \rightarrow 0$, as $\rho \rightarrow 0^+$ or $\rho \rightarrow \infty$. Furthermore, the important maximum uncertainty point $\rho_{M,max}$, at which entropy $S(\rho)$ has a maximum value, is a solution of the governing equation (obtained as $S'(\rho) = 0$):

$$\sum_{n=0}^M [n - \bar{N}(\rho)] \cdot (a_n \cdot \rho^n) \cdot \ln(a_n \cdot \rho^n) = 0. \quad (6)$$

It can be noted that if $M \rightarrow \infty$, then entropy S is a monotonically unlimited function of the parameter ρ (then, $\rho < 1$). A more detailed analysis of the mutual change of the random variable N and its carried information i_N has been provided in the study [24].

2.2 Binomial-Type System

This study models the binomial system as a family of birth-death processes of size $M + 1$ and transition rates as follows:

$$\lambda_n = \lambda \cdot (M - n), \quad \mu_{n+1} = (n + 1) \cdot \mu, \quad n = 0, 1, \dots, M - 1. \quad (7)$$

The following equations can be derived:

$$\frac{p_n(\rho)}{p_0(\rho)} = \binom{M}{n} \cdot \rho^n, \quad a_n = \binom{M}{n}, \quad n = 1, 2, \dots, M,$$

$$f_M(\rho) \equiv 1 + \sum_{n=1}^M \binom{M}{n} \cdot \rho^n = (1 + \rho)^M, \quad \rho > 0, \quad M > 1.$$

Thus,

$$p_n(\rho) = \frac{\binom{M}{n} \cdot \rho^n}{f_M(\rho)} = \frac{\binom{M}{n} \cdot \rho^n}{(1 + \rho)^M}, \quad n = 0, 1, \dots, M; \quad \rho > 0. \quad (8)$$

Therefore, the quantity N obeys the binomial distribution of size $M + 1$ and parameter ρ , with $\rho > 0$. In addition, through $p = \rho/(1 + \rho)$, with $p \in (0, 1)$, the standard form of the binomial distribution family can be obtained as follows:

$$p_n(\rho) = \text{Prob}\{N = n\} = \binom{M}{n} \cdot p^n \cdot (1 - p)^{M-n}, \quad n = 0, 1, \dots, M.$$

The following equation can be derived from Eq. (4):

$$\bar{N}(\rho) = \frac{M \cdot \rho}{1 + \rho}, \quad \rho \in (0, +\infty), \quad \bar{N}(p) = M \cdot p, \quad p \in (0, 1). \quad (9)$$

The mean $\bar{N}$ for the binomial system for $M = 5, 10, 15$, is depicted in Figures 1 and 2.

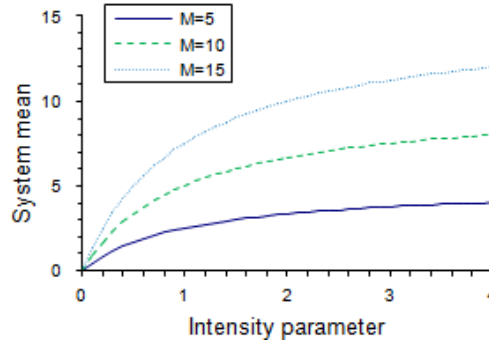


Figure 1. System mean $\bar{N}$ vs. intensity parameter ρ for the binomial system

Note: $\bar{N} = M/2$ for $\rho = \rho_{M,max} = 1$.

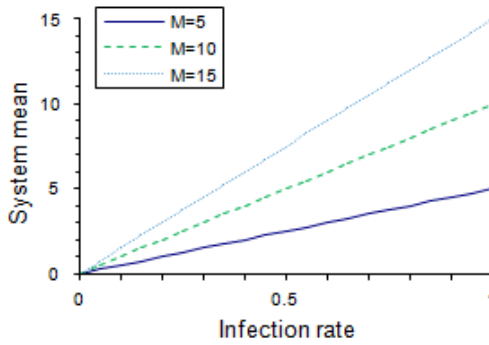


Figure 2. System mean $\bar{N}$ vs. infection rate p for the binomial system

Note: $\bar{N} = M/2$ for $p = p_{M,max} = 0.5$.

For the entropy of the binomial system, the following equation can be derived from Eqs. (2), (3), and (8):

$$S(\rho) = M \cdot \left[\ln(1 + \rho) - \frac{\rho \cdot \ln \rho}{1 + \rho} \right] - \frac{1}{(1 + \rho)^M} \cdot \sum_{n=1}^M \binom{M}{n} \cdot \ln \binom{M}{n} \cdot \rho^n, \quad \rho > 0. \quad (10)$$

Eq. (10) has also been used (e.g. [25]). The entropy $S(\rho)$ for the binomial system for $M = 5, 10, 15$, is depicted in Figure 3. This study presents the entropy in the following form:

$$S(p) = -M \cdot [p \cdot \ln p + (1 - p) \cdot \ln(1 - p)] - \sum_{n=0}^M \binom{M}{n} \cdot \ln \binom{M}{n} \cdot p^n \cdot (1 - p)^{M-n}, \quad p \in (0, 1). \quad (11)$$

The entropy $S(p)$ for the binomial system for $M = 5, 10, 15$, is also presented in Figure 4.

According to Eq. (6), for any field size M , the maximum uncertainty point $\rho_{M,max} = 1$. Therefore, using the above notation, $p_{M,max} = \rho_{M,max} / (1 + \rho_{M,max})$ and $p_{M,max} = 0.5$ can be obtained. So, after Eqs. (10) and (11), $S_{max} = S(\rho_{M,max} = 1) = S(p_{M,max} = 0.5)$, i.e.,

$$S_{max} = M \cdot \ln 2 - \frac{1}{2^M} \cdot \sum_{n=1}^M \binom{M}{n} \cdot \ln \binom{M}{n}, M \geq 1.$$

In addition, $\bar{N} = M/2$ can be obtained through Eq. (9). The binomial system of size M is a symmetrical one in information sense, with respect to the infection intensity point $\rho = 1$, that is, $S(\rho') = S(\rho'')$ for any pair $\rho' \cdot \rho'' = 1$; with respect to the infection rate point $p = 0.5$, that is, $S(p') = S(p'')$ for any pair $p' + p'' = 1$. Table 1 shows the parameters and measures defined in this section.

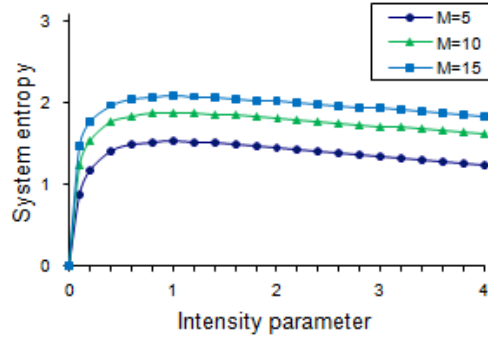


Figure 3. System entropy S vs. intensity parameter ρ for the binomial system

Note: $\rho_{M,max} = 1$ for any $M \geq 1$.

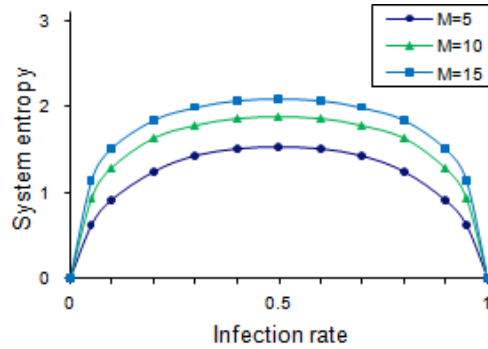


Figure 4. System entropy S vs. infection rate p for the binomial system

Note: $p_{M,max} = 0.5$ for any $M \geq 1$.

Table 1. Input parameters and output measures of the plant field model

Notation	Explanation	Formula
λ	Rate at which an infected plant infects a susceptible plant	—
μ	Rate at which an infected plant recovers or is removed	—
N	Current number of infected field plants	—
M	Maximum number of field plants, i.e., the field size	—
ρ	Field infection intensity	$\rho = \lambda/\mu$
p	Field infection rate	$p = \rho/(1 + \rho)$
$\bar{N}$	Average number of infected field plants	Eq. (9)
S	Field entropy	Eqs. (10) and (11)

Note: “—” indicates not applicable.

2.3 Entropy as an Operational Measure

This study addresses a methodology for managing uncertainty in agriculture business operations and strategic planning for decision support. Therefore, this research is designed to help farm managers and agribusiness executives make precise, data-driven decisions on when and where to allocate their resources. As a decision rule, this study proposes an entropy-based inspection through the field infection intensity ρ , i.e., the field infection rate p . For a considered plant field of size M , the variable N denotes the current number of infected plants in the field. This study experimentally measures this quantity and calculates the parameter ρ . Then, using Eqs. (10) and (11), the concrete entropy value is obtained (Figures 3 and 4).

This study suggests a window of medium volatility (e.g., $p = \bar{N}/M \in (0.3; 0.7)$), where an inspection becomes necessary, and a window of maximum volatility (e.g., $p = \bar{N}/M \in (0.4; 0.6)$), where an intervention dictates whether the field accelerates into collapse or stalls. The highest sensitivity to perturbation ultimately maximizes the economic efficiency of allocated crop protection assets. In both cases, the point $p_{M,max} = 0.5$ divides the interval into two equal parts, where the left portion signals field recovery, while the right portion indicates a transition toward full-field infection.

3 Modeling of Pathogen Spread in a Greenhouse Tomato Plot

As an illustrative example, this study considers an agro-statistical trial conducted on an isolated greenhouse plot, each containing exactly M tomato plants. Because greenhouse settings isolate the crops from open-air migration, the plant population functions as a strictly closed system, fulfilling the criteria of a finite binomial birth-death process. Over several weeks, visual inspections can be conducted weekly to monitor the exact count of infected individuals N per plot, aiming to calculate the infection rate p . This study can also measure the infection (birth) rate per week and the removal (death/pruning) rate per week, aiming to calculate the infection intensity as $\rho = \lambda/\mu$.

This study calculates the average number of infected plants at any given time through Eq. (9) and the entropy of the agricultural field through Eqs. (10) and (11). Considering some simulated observations, for $M = 5, 10, 15$ and taking $\rho = 1/2; 3/4; 1; 4/3; 2$ (i.e., $p = 1/3; 3/7; 0.5; 4/7; 2/3$), the related results can be found in Tables 2–4. This study assumes that the input parameter ρ (i.e., p) is progressing, corresponding to an infection spreading from individual plants to the entire field, where the entropy model identifies the exact threshold for intervention. This study considers three stages for a 60-day period as follows:

- Early stage (days: 0–15): Disease presence is localized. The entropy is low because the system is predictably healthy.
- The critical stage (around day 30): When exactly half the field is infected (approximately $M/2$), the entropy peaks at the maximum uncertainty point. At this precise moment, a single plant's transition dictates whether the plot accelerates into complete collapse or stalls.
- Late stage (days: 45–60): The pathogen saturates the system. The entropy falls back toward zero as complete infection becomes a deterministic certainty.

Table 2. Output measures for a plant field with field size $M = 5$ for different values of λ and μ

λ	μ	ρ	p	$\bar{N}$	S	$\bar{N}/M$	S/S_{max}
2	4	0.5	0.33	1.65	1.446	33%	94.9%
3	4	0.75	0.43	2.15	1.511	43%	99.1%
3	3	1	0.5	2.5	1.524	50%	100%
4	3	1.33	0.57	2.85	1.511	57%	99.1%
4	2	2	0.67	3.35	1.446	67%	94.9%

Table 3. Output measures for a plant field with field size $M = 10$ for different values of λ and μ

λ	μ	ρ	p	$\bar{N}$	S	$\bar{N}/M$	S/S_{max}
2	4	0.5	0.33	3.3	1.808	33%	96.4%
3	4	0.75	0.43	4.3	1.865	43%	99.4%
3	3	1	0.5	5.0	1.876	50%	100%
4	3	1.33	0.57	5.7	1.865	57%	99.4%
4	2	2	0.67	6.7	1.808	67%	96.4%

The assumed 60-day disease progression period is connected to the numerical values in Tables 2–4, in a way that the parameter ρ (i.e., p) increases with the spread of the infection, starting from individual plants and eventually

covering the entire field. There, the first and fifth rows correspond to the early and late transitional case, respectively, while the third row represents the critical stage, with the second and fourth rows acting as intermediate transitional cases.

Regarding the impact of agricultural policies, this case study suggests that evaluating an agricultural field as a binomial birth-death process allows agronomists to calculate the critical window of maximum volatility. Applying bio-fungicides on or before day 30 targets the system when it has the highest sensitivity to perturbation, maximizing the economic efficiency of crop protection assets.

Table 4. Output measures for a plant field with field size $M = 15$ for different values of λ and μ

λ	μ	ρ	p	$\bar{N}$	S	$\bar{N}/M$	S/S_{max}
2	4	0.5	0.33	4.95	2.014	33%	96.9%
3	4	0.75	0.43	6.45	2.069	43%	99.5%
3	3	1	0.5	7.5	2.079	50%	100%
4	3	1.33	0.57	8.55	2.069	57%	99.5%
4	2	2	0.67	10.05	2.014	67%	96.9%

3.1 Discussion on Symmetric Balances

Considering two inverse symmetrical points of the infection intensity ρ with respect to the maximum uncertainty point $\rho_{M,max} = 1$ (i.e., $\rho' \cdot \rho'' = 1$), or of the infection rate p with respect to the maximum uncertainty point $p_{M,max} = 0.5$ (i.e., $p' + p'' = 1$), then this study confirms (Tables 2–4):

$$\bar{N}(\rho') + \bar{N}(\rho'') = M, \text{ or } \bar{N}(p') + \bar{N}(p'') = M,$$

$$S(\rho') = S(\rho''), \text{ or } S(p') = S(p'').$$

Thus, the entropy S is an inverse symmetrical function of the parameter ρ (i.e., p), and the mean $\bar{N}$ is a complementary function to size M of the parameter ρ (i.e., p). Therefore, any recovery conditions are anticipated to mitigate the field mean infection $\bar{N}$, and it is essential to apply them for higher values of entropy S , where the risk of full-field transmission escalates. This study considers a plant field of size M for the presence of a destructive beetle. It can be seen that the normalized ratio $\bar{N}/M$, being actually the infection rate p , stays the same as M increases, but the normalized ratio S/S_{max} also increases as M increases (for a given infection rate p).

Remark 1. For a sample consisting of m observations for the current infected plants N , if the estimation of the infection intensity parameter ρ is a concern, then the entropy-based estimation of the parameter ρ should be used through Eqs. (10)–(11), instead of the maximum likelihood estimation (which is based on the average number of infected plants).

4 Conclusion

This study demonstrates that integrating information theory with stochastic population dynamics provides a robust framework for quantifying agricultural risk. By modeling a closed greenhouse plot as a finite binomial birth-death process, this study successfully mapped the pathogen trajectory not just by its physical spread, but through its structural uncertainty via its field entropy. The findings confirm that the pathogen spread uncertainty peaks precisely at the 50% infection threshold, where entropy reaches its global maximum. Beyond this tipping point, system uncertainty collapses as the infection process approaches full-field saturation. Linking this entropy curve to a dynamic mean loss function provides that information-theoretic metrics could serve as reliable future metrics in optimizing both real-time pathogen containment and long-term capital allocation in precision agribusiness.

The analysis presented in this study transitions the entropy from an abstract statistical concept into an actionable decision-making tool for modern agriculture. Operationally, the inverted-U-shaped/unimodal form of the entropy function provides farm managers with an explicit “window of maximum volatility”. Rather than relying on rigid calendar-based spraying or wait-and-see protocols, triggering interventions based on real-time sample entropy peaks maximizes the efficiency of pesticide assets and preserves healthy yields before the outbreak becomes deterministic. Strategically, integrating this birth-death framework into corporate asset planning allows agribusiness executives to calculate the uncertainty (i.e., the entropy) associated with any plant field. The entropic threshold provides farmers with actionable insights, knowing exactly when to apply and when to withhold pesticides to maximize crop resilience.

Data Availability

The data supporting our research results are included within the article.

Conflicts of Interest

The author declares no conflicts of interest.

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